



Initial marine movements of silver European eels *Anguilla anguilla* following their emigration from an English chalk stream

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Abstract The conservation management of the facultative catadromous and critically endangered European eel *Anguilla anguilla* is inhibited by substantial knowledge gaps on their behaviours and migrations. Integral to understanding the silver eel migration back to the Sargasso Sea for spawning is generating knowledge on their initial marine movements. Here, the downstream migration of 25 silver eels from the River Test, southern England, and their initial

movements in the sea were measured using acoustic telemetry. Of 24 eels detected migrating downstream, 19 were successfully tracked into the tidal area (taking 1–34 days), with in-river barriers not delaying their emigration. Thirteen eels were then detected in the marine environment, with 11 immediately moving westwards towards the Atlantic Ocean. These 11 eels took up to 25 days to move to areas outside of the marine receiver array, a linear distance of approximately 80 km from the river mouth. While eels were generally detected moving downstream in the river during darkness, marine detections occurred across the 24-h cycle, with detection depths up to 50 m. These movement data indicated eels generally migrated relatively quickly out of the river, with most then moving through inshore areas in the direction of their spawning area.

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Introduction

The migration timing and success of diadromous fishes is subject to natural selection pressures, including pressures imposed by temperature, precipitation and photoperiod regimes (La Sorte et al., 2016). Artificial selection pressures can also act on their migrations, especially the presence of anthropogenic riverine barriers and artificial side channels

that individuals must navigate over or through if they are to complete their migration (Duarte et al., 2021; Verhelst et al., 2021). During their emigration from freshwater to marine environments, the factors that trigger migration and influence its success are thus important to determine so that their escapement rates can be better understood (Sigourney et al., 2015). Once into the marine environment then assessing the initial movements of successful migrants, including their movement direction and speed, provides information on their and initial behaviours and migration pathways towards their spawning and/or foraging areas (Verhelst et al., 2022; Barry et al., 2026).

In critically endangered European eel *Anguilla anguilla* (Linnaeus, 1758) (“eel”), the continental phase of their lifecycle concludes when individuals metamorphose from yellow to silver eels. The biological distinction between the sedentary yellow stage and the migratory silver stage is driven by physiological modifications that enable their switch from river residency to marine migration (Durif et al., 2005; Righton et al., 2016). Female eels tend to metamorphose to silvers at relatively large body sizes, whereas males tend to be smaller and younger (Helfman et al., 1987; Aprahamian et al., 2007). The initiation of silver eel emigration from European freshwaters back to the Sargasso Sea (Wright et al., 2022) tends to peak in autumn and spring (Aarestrup et al., 2008; Lagarde et al., 2023), with autumn migration generally predominant at higher latitudes (Verhelst et al., 2025).

The phenology of the downstream emigration of silver eels has been associated with multiple environmental triggers, including hydrological variables (especially elevated discharge and flow velocity) and changes in barometric pressure (e.g. Sandlund et al., 2017; Monteiro et al., 2020). Increased migration activity generally occurs in the less-illuminated phases of the lunar cycle (Barry et al., 2016; Bašić et al., 2019), although some studies have also suggested a negligible influence of moon phase (e.g. Reckordt et al., 2014). However, controlled experiments on silver eels have suggested a direct avoidance of artificial light (e.g. Cullen and McCarthy, 2000) and a preference for distinct nocturnal behaviour (Riley et al., 2011), suggesting that emigration is more likely on darker nights. The time taken for silver eels to emigrate from rivers is also important, with migration delays incurred at barriers increasing mortality risk (e.g. from predation) (Trancart et al., 2020). The threat of injury or death of individual eels

is elevated further when these barriers are hydroelectric plants or flow regulatory structures (e.g. pumping stations), where the primary passage route is through the turbines or pumps (McCarthy et al., 2008; Bolland et al., 2019).

A recent meta-analysis on silver eel emigration covering 18 European rivers indicated that silver eel phenology is highly plastic, with variability apparent in their migration speeds and the timing of arrival of migrants into the marine environment (Verhelst et al., 2025). Moreover, once silver eels arrive in the marine environment, they do not always immediately move in the direction of the Sargasso Sea. Instead, some will remain in coastal areas for short periods, but with this marine residency continuing for a number of years in some individuals (Vøllestad et al., 1994; Aarestrup et al., 2008). There is some evidence that the movement of freshwater resident eels from river into sea is reversible, with some silver eels returning to their river of origin after spending up to a year in the marine environment, although their stay in the freshwater environment is generally short (<33 days; Tambets et al., 2021). Once their actual marine migration to the Sargasso Sea commences, then the routes from northern Europe include a ‘Nordic’ route from Scandinavia to the north of the British Isles into the Atlantic, a route from the west coast of Ireland towards the Azores, and one through the English Channel (Verhelst et al., 2022).

The aim here was to apply acoustic telemetry to track the riverine emigration of silver eel from a chalk stream in Southern England—the River Test—and into their inshore areas of the English Channel. The first objective was to determine the proportion of tagged silver eels leaving the river and the proportion subsequently detected in the inshore marine environment, while assessing how instream barriers and environmental drivers influence their downstream progression. The second objective was to assess the temporal and spatial patterns of eel detections in the marine environment to inform their initial offshore movements.

Materials and methods

Study river, eel tagging and the acoustic receiver array

Chalk streams are aquifer fed, leading to relatively stable temperatures and hydrology, and are generally

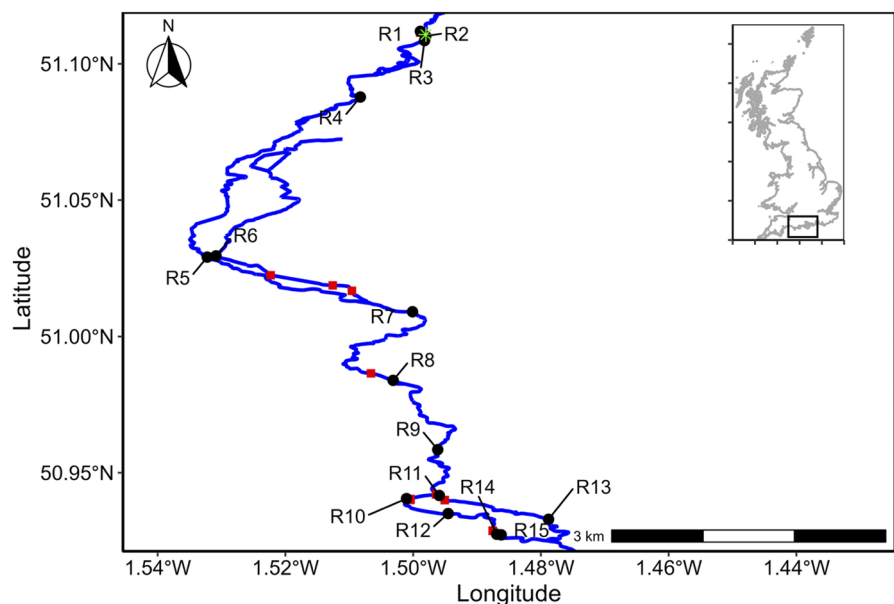
highly productive riverine ecosystems (Simmons et al., 2021). The River Test is such a chalk stream. It flows for 139 km from its source (51.2434, −1.2393) to its confluence with the English Channel (50.8221, −1.2880). This study focused on the downstream reaches of the river below Stockbridge (51.1102, −1.4980), with the river length to the estuary being approximately 25 km. This section of river does braided channels present in some areas (Fig. 1). River widths are generally up to 20 m and depths to 2 m. The river drains into Southampton Water, which is an approximately 10-km-long and 2-km-wide extension of the Central Solent (Fig. 2). The immediate inshore area is then dominated by the Isle of Wight, with Chichester Harbour to the east, Christchurch and Poole Harbours to the west, with the western most landmark of the study area being the Isle of Portland (Fig. 2). These inshore areas are generally shallow (<20 m) but with some areas of deeper water off the Isle of Wight (>100 m).

A total of 25 silver eels were captured by commercial fishermen using an eel trap on the River Test in Stockbridge between September and October 2023. (The exact dates of capture were unavailable.) The captured eels then held in a trout fish farm facility before tagging on 09/11/2023 (mean total length $\pm$ SE: 676 ± 12 mm). Tagging involved anaesthetising individual eels with 2-phenoxyethanol (2 ml l^{-1}) followed by the gastric insertion of a coded acoustic

transmitter with a temperature and pressure sensor using a 5-mm-diameter glass insertion rod (transmitter model: V9TP, 69 kHz, Innovasea; 30-s delay for first 20 days, 50-s delay thereafter, at high power) (Økland and Thorstad, 2013). This programming was designed to capture the initial silver eel movements through the river at a relatively high resolution but with this resolution then reduced as the eels moved into the marine environment, although the results suggested the eels behaved differently to this (*cf.* Tables 1, 2). The pressure sensor on the transmitters had an accuracy of ± 1 m and resolution of ± 0.3 m at a maximum depth of 68 m. Each eel was then externally tagged with a T-bar anchor “do not eat” tag (30-mm marker + 20-mm anchor). Prior to tagging, eels were weighed (to 0.1 g) and measured for total length (nearest mm), left pectoral fin size and vertical and horizontal left eye diameter. After their recovery to normal behaviour (e.g. maintenance of body equilibrium, responding to external stimuli), the eels were released into the adjacent river Sect. (51.1112; −1.4980). All tagging procedures were carried out under a UK Home Office Project License (PPL P3BF61EDF) under the Animals (Scientific Procedures) Act 1986 and following ethical review and approval.

The movements of the tagged eels were monitored using 15×69 kHz Innovasea VR2W acoustic receivers strategically positioned in the River Test (one

Fig. 1 Receiver locations along the River Test. Black circles show the locations of each receiver and their code (as R1 to R15; *cf.* Table 1), red squares are in-river barriers, and the green star is the release site of the tagged eels



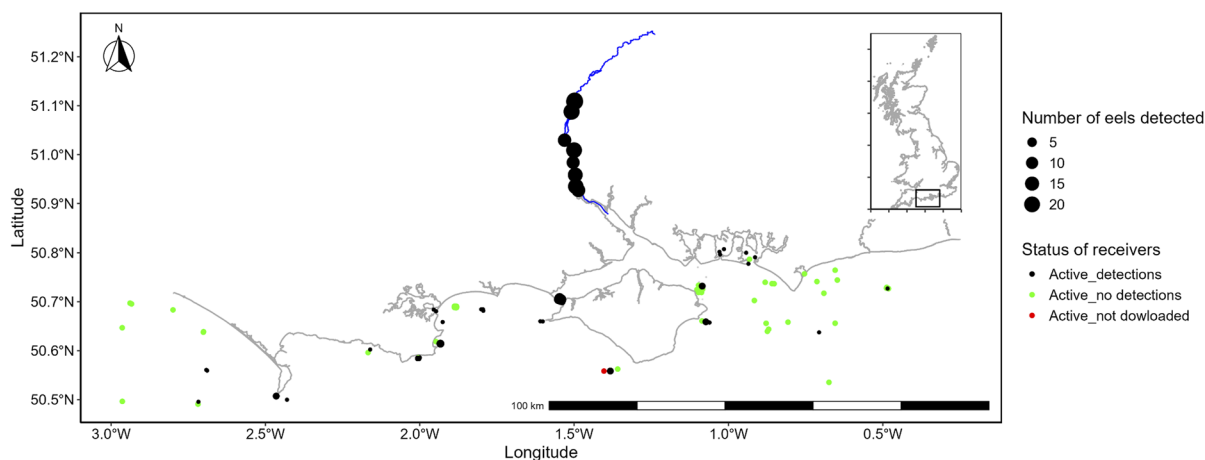


Fig. 2 Receiver locations along the River Test and in the English Channel. Black circles show eels detected at each receiver with diameter corresponding with the number of individuals detected. Green circles denote active receiver with no detec-

tions and red circles active receivers whose information has not been downloaded. For eels detected in the English Channel, the first date of detection on each receiver is available on Fig. S2

Table 1 Locations of in-river acoustic receivers, the distance downstream of the release site (where eels were detected) and the total number of detections and eels detected per receiver

Receiver	Location	Distance down-stream (m)	<i>n</i> detections	<i>n</i> eels
R1	51.11195, −1.4988667	n/a	0	0
R2	51.1102, −1.4980333	100	875	22
R3	51.10855, −1.4982167	300	105	24
R4	51.087817, −1.5082667	2912	88	21
R5	51.02905, −1.5322333	10,278	1	1
R6	51.029483, −1.5308667	10,219	2152	13
R7	51.008983, −1.5001	13,546	170	20
R8	50.983917, −1.5031333	16,823	14	12
R9	50.958483, −1.49615	20,012	30	17
R10	50.9405, −1.5010333	22,259	1	1
R11	50.94165, −1.4959	18,945	0	0
R12	50.935017, −1.4945167	23,083	61	19
R13	50.932917, −1.4788	19,900	0	0
R14	50.927417, −1.4869167	24,048	22	9
R15	50.9272, −1.4862167	24,048	32	13

receiver upstream of the release site, and 14 downstream, including coverage of braided channels in the lower river; Table 1, Fig. 1). There was then an array of 91 receivers located across the inshore area (combination of 69 kHz VR2AR and VR2TX) at sea as part of the Fish Intel Network (Hall et al., 2025), with the tagged eels detected on 33 of these receivers. These receivers were placed into strategic areas around the onshore areas of the Isle of Wight, to the

east in Chichester Harbour and through areas to the west of the Isle of Portland (Fig. 2; Supplementary material, Table S1). This receiver spatial coverage thus enabled the tracking of any eels that moved west through the English Channel and towards the Atlantic Ocean, as well as any that stayed resident within the Solent area (Fig. 2; Table S1).

Field logistics prevented receiver ranges being calculated across the study river. However, the receiver

Table 2 Transmitter serial numbers, eel lengths, weights and silvering stage (cf. Durif et al., 2005), and progression rates in the river and marine environment. Note that the transmitters had a 30-s delay for the initial 20 days post-tagging but then switched to a 50-s delay thereafter. For eels detected in the English Channel, the first date of detection on each receiver is available in Fig. S2

Transmitter	L (mm)	W (g)	Silvering stage	<i>n</i> receivers detected in river	Detected in lower river (R12-R15)	Progression rate in river (m s^{-1})	<i>n</i> receivers detected in sea	Date of first detection	Date of last detection in river	Date of last detection at sea	Linear distance tracked (m)	Progression rate in marine movement (m s^{-1})
1,577,582	650	498.9	Yellow/Pre-migrant	10	Y	0.71	12	09/11/2023	14/11/2023	18/11/2023	104,850	0.46
1,577,580	706	582	Yellow/Pre-migrant	10	Y	0.01	10	09/11/2023	09/12/2023	13/03/2024	51,820	0.002
1,577,574	581	395.1	Yellow/Resident	10	Y	0.07	1	09/11/2023	14/11/2023	06/12/2023	66,762	0.03
1,577,566	595	449.9	Yellow/Resident	10	Y	0.07	0	09/11/2023	13/11/2023	n/a	23,083	n/a
1,577,573	703	647.1	Yellow/Pre-migrant	10	Y	0.02	0	09/11/2023	07/12/2023	n/a	23,948	n/a
1,577,564	703	587.9	Yellow/Pre-migrant	9	Y	0.66	9	09/11/2023	10/11/2023	04/12/2023	84,600	0.04
1,577,577	628	447.3	Yellow/Resident	9	Y	0.71	4	14/11/2023	15/11/2023	24/11/2023	66,811	0.003
1,577,557	601	314.2	Silver/Migrant	9	Y	0.21	0	15/11/2023	15/11/2023	n/a	24,048	n/a
1,577,575	812	960.2	Silver/Migrant	8	Y	0.01	6	14/11/2023	04/12/2023	20/02/2024	62,260	0.005
1,577,569	697	569.9	Yellow/Pre-migrant	8	Y	0.05	2	11/12/2023	19/12/2023	11/01/2024	66,762	1.16
1,577,565	781	982.6	Yellow/Pre-migrant	8	Y	0.01	0	09/11/2023	04/12/2023	n/a	24,048	n/a
1,577,562	683	585.6	Yellow/Pre-migrant	8	Y	0.30	3	09/11/2023	05/12/2023	11/12/2023	62,300	0.48
1,577,559	748	717.8	Yellow/Pre-migrant	8	Y	0.52	0	14/11/2023	05/12/2023	n/a	23,083	n/a
1,577,570	645	438.9	Yellow/Resident	7	Y	0.01	2	09/11/2023	04/12/2023	07/12/2023	69,050	n/a
1,577,571	640	502.2	Yellow/Resident	7	Y	0.01	1	21/11/2023	08/12/2023	13/12/2023	68,930	0.12
1,577,572	610	402.6	Yellow/Pre-migrant	7	Y	0.01	1	10/11/2023	09/12/2023	13/12/2023	24,048	0.18

Table 2 (continued)

Transmitter	L (mm)	W (g)	Silvering stage	<i>n</i> receivers detected in river	Detected in lower river (R12-R15)	Progression rate in river (m s^{-1})	<i>n</i> receivers detected in sea	Date of first detection	Date of last detection in river	Date of last detection at sea	Linear distance tracked (m)	Progression rate in marine movement (m s^{-1})
1,577,560	681	669.8	Silver/Migrant	7	Y	0.02	3	19/11/2023	04/12/2023	08/12/2023	66,762	0.02
1,577,578	683	616.3	Yellow/Pre-migrant	7	Y	0.01	3	09/11/2023	13/12/2023	21/01/2024	94,820	0.59
1,577,567	730	654.7	Yellow/Pre-migrant	5	Y	0.01	0	12/11/2023	04/12/2023	n/a	13,546	n/a
1,577,568	700	830.8	Silver/Pre-migrant	4	N	0.002	0	09/11/2023	03/01/2024	n/a	13,446	n/a
1,577,561	574	357.5	Yellow/Pre-migrant	3	N	0.06	0	10/11/2023	10/11/2023	n/a	2912	n/a
1,577,579	662	40	Yellow/Resident	2	N	0.37	0	09/11/2023	09/11/2023	n/a	300	n/a
1,577,558	683	717.4	Silver/Pre-migrant	2	N	0.002	0	12/11/2023	14/11/2023	n/a	100	n/a
1,577,576	774	848.2	Yellow/Pre-migrant	2	N	0.05	0	09/11/2023	09/11/2023	n/a	300	n/a
1,577,558	683	717.4	Silver/Pre-migrant	2	N	0.0008	0	12/11/2023	14/11/2023	n/a	99	n/a

coverage was considered sufficient for tracking eels from the release site to the sea, based on the combined effect of the tagging protocol (using short transmitter delays at high power) and the study river's relatively stable, groundwater-fed flow regime, which prevented rapid increases in discharge (Figs. 1, 2). Post hoc receiver efficiencies were calculated as ratios of eels detected at a specific receiver against the total number of eels subsequently recorded at downstream receivers. The distances between the receivers were calculated using open-source scripts developed by the Research Institute for Nature and Forest (INBO), available at their GitHub repository (https://github.com/inbo/fish-tracking/tree/master/scripts/receiver_distance_analysis).

Data analyses

Each eel was assigned a silvering stage using the Durif silvering index (Durif et al., 2005) based on body weight, total length and measurements of the left pectoral fin and left eye mean diameter. The total number of receivers detecting at least one eel was determined, followed by the number of detected eels per receiver and the total number of detections per receiver. Receiver locations were classified as either river (in fresh water) or sea (in brackish or salt water). For an eel to have been interpreted as exiting the river, it had to be detected on at least one of the most downstream riverine receivers, R12, R14 and R15, with R14 and R15 considered to be at the tidal limit, and R12 approximately 1 km upstream (Table 1). All statistical analysis and graphical outputs were performed using R version 4.4.3 (R Core Team, 2022).

Progression rates were calculated as the distance travelled divided by time for eels (i) within the river (as first to last detection in the river); (ii) moving between the river to sea (as first detection in the river to first detection in the sea); and (iii) from the river to final detection (as the first detection in river to final detection at sea). Distances travelled were calculated for each eel by summing the total distance moved between receivers, with the receiver detection ranges considered not to overlap due to their distances separating them (Table 1). To assess the influence of barriers on movement downstream, the river was separated into reaches impacted by barriers and reaches that were free flowing, with these also related to receiver location. Barrier data were

obtained from the Environment Agency's River Obstacles dataset (<https://river-obstacles-theriverstrust.hub.arcgis.com/>). For each reach, the rate of progression was calculated from the time of the first detection at the most upstream receiver to the time of the final detection at the most downstream receiver within that reach. The time taken was then expressed in relation to distance (m s^{-1}). Differences in progression rates between reaches with and without barriers were tested in a linear mixed-effects model completed in the lme4 R package version 4.4.3 (Bates, 2010). The data were initially inspected for missing values or irregularities, and the model fitted using the *lmer* function. The full model had progression rate as the dependent variable and eel ID as a random effect. Fixed factors were eel length (standardised to mean eel length (0) and individual lengths expressed as the number of standard deviations away from the mean), silvering stage (expressed as a factor represented the stage, where a higher factor represented a more advanced silvering stage), number of barriers, moon illumination, silvering stage, river flow, tide phase (categorical) and water temperature. Moon illumination was obtained as a fraction using the *getMoonillumination* function from the *suncalc* package version 0.5.1 (Thiermul et al., 2019), varying from 0 at new moon to 1 during a full moon. River flow was taken from the Department for Environment Food & Rural Affairs hydrology long-term records, recorded every 15 min during the study period at Timsbury (51.008915, -1.5002201, <https://environment.data.gov.uk/hydrology/explore>). River temperature was continuously monitored every 15 min throughout the study period using temperature loggers (Gemini Tinytag Aquatic 2, TG-4100, range -40 to +70 °C; measurement accuracy of ± 0.5 °C; reading resolution of 0.01 °C) co-located with five specific receivers along the river (R2, R5, R7, R12, R14). Tide height values for Portsmouth were obtained from the UK Tide Gauge Network (British Oceanographic Society, 2024). To assign each tidal observation to a phase of the tidal cycle, we first identified the reference high tide by locating the maximum tidal height within the dataset. For each observation, we computed the time difference (in hours) from the reference high tide and assigned either high, ebb, low or rising tide based on a tidal cycle length of 12.42h, with 25% of continuous 15-min intervals assigned to each category (Fig. S1). For categorical variables, full pairwise comparisons

were computed using the emmeans package version 1.11.2 (Lenth et al., 2025).

Results

Riverine detections, emigration rates and movements

There were 3570 eel detections recorded across 12 riverine receivers, with three receivers not detecting any eels, including the receiver upstream of the release site (Table 1). The detection efficiencies of the riverine receivers were highly variable, with some receivers having relatively high efficiencies (R3: 100%, R2: 92%) but others were either highly inefficient (e.g. R14: 36%) or located on side channels through which no eels moved through (e.g. R13, 0%) (Table 1).

Of the 25 tagged eels, one eel was never recorded, and 24 were detected as initially moving downstream (on R3), with none detected moving upstream (on R1) (Table 1; Figs. 1, 2). There were then 19 of the eels detected on R12, 23.2 km from R2, of which nine were detected within the tidal limit on R14 and 14 were also detected on R15 (Table 1; Figs. 1, 2). Eel total length did not differ significantly individuals detected on R12 to R15 and those that did not reach R12 (Welch two-sample *t*-test: $t = 0.87$, $df = 10.7$, $P = 0.41$). The mean length of eels that were tracked downstream to at least R12 was 678 ± 28 mm ($n = 19$) versus 679 ± 63 mm ($n = 5$) of those eels detected as moving but not reaching the lower river (Figs. 1, 2; Table 2). The time between the first detection in the river and their final detection in the lower

river was between 1 and 34 days (median 20 days (IQR: 5–25 days)).

Eel arrivals at downstream receivers occurred predominantly at night (Table 3), with 98.5% of river detections (129 of 131) and 72.1% of sea detections (75 of 104) recorded during the night. Most eels started moving downstream soon after release (21 of 25 were first detected between 9 and 15th November; Table 2). Eel arrival at downstream receivers varied according to the tidal cycle (Fig. 3). In the river, eels were most likely to arrive at a downstream receiver during a rising tide (n observations = 52; 39.7%) or high tide (n observations = 37; 28.2%), while in the sea, arrival at new receivers during an ebb tide was most common (n observations = 44, 42.3%). Progression rates through the river varied by individual eel, with a maximum rate of 0.71 m s^{-1} (Table 2).

We used a linear mixed-effects model to examine the effects of environmental and biological predictors on eel speed (as progression downstream). To evaluate the contribution of each variable, model selection was conducted by comparing the full model with reduced models using AICc, using the MuMIn package version 1.48.19 (Bartoń, 2026). Removing tide phase and eel length significantly improved model fit (ΔAICc six and 15, respectively), and these predictors were therefore excluded from the final model (Table S2), despite the pattern seen in tidal arrivals. In contrast, removal of moon illumination and silvering stage significantly reduced model fit, and these were retained in the final model which included temperature, barriers, flow, moon illumination and silvering stage as fixed effects (Table 3). The model indicated that eels progressed downstream more quickly at lower temperatures and during periods of lower

Table 3 Results of the linear mixed-effects model testing the influence of environmental variables, barriers (weirs, sluices, dams) and silvering stage of the eels on their progression rate

Term	Estimate	Std. Error	<i>t</i> -value	<i>P</i> -value
(Intercept)	1.414	1.078	1.312	0.193
Temperature	−0.144	0.07	−2.064	0.042*
Barriers	0.082	0.038	2.128	0.037*
Flow	0.047	0.025	1.883	0.063
Moon fraction	−0.742	0.238	−3.114	0.003**
Silvering stage female pre-migrant	−0.201	0.18	−1.118	0.284
Silvering stage female resident	−0.13	0.2	−0.65	0.527

Significance levels are indicated by asterisks, where *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$

downstream, with progression rate as the dependant variable and eel ID as a random effect. Bold values indicate statistically significant factors at 95% CI

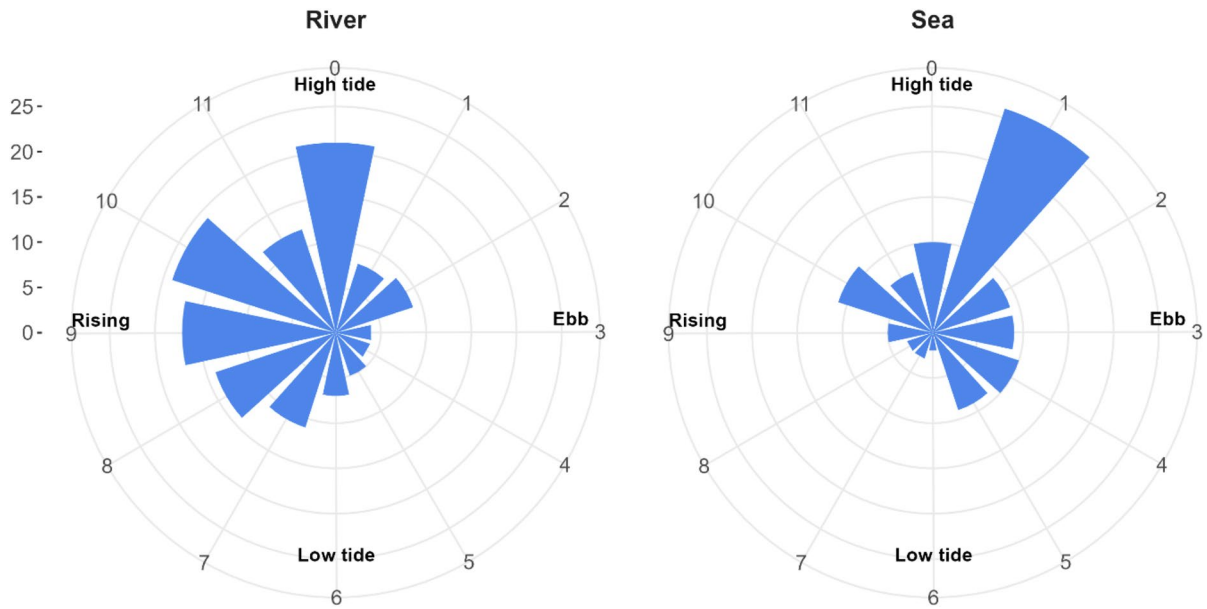


Fig. 3 Eel arrivals at downstream receivers in both the river and the sea based on the tidal cycle. The dashes on the left represent the number of observations. Data were pooled from all 25 eels

moon illumination and also moved more quickly through sections impacted by barriers (Table 3). River flow did not have a significant effect (Table 3). Full pairwise comparisons between the speed of different silvering stages are provided in Table S3.

Marine detections and movements

Of the 19 eels detected on R12 (of which 15 were detected on R14 and/or R15), 13 were then detected on 33 of the marine receivers, with a total of 746 detections (Table S1, Figs. S2), where the number of days between their final detection in the river and their final detection in the sea ranged between 3 and 95 days (median (IQR): 9 (4–24) days) (Table 2). Only one eel was still being detected at the approximate period of transmitter expiry, which was one of the eels detected as moving eastwards, with this eel entering Chichester Harbour and not moving thereafter (Figs. 2, S3). Other eels were all detected as moving westwards (although two initially moved eastwards before being detected to the west of the river mouth). Individual marine receivers detected up to 7 tagged individuals (Western gate CW), and eels being detected for up to 75 days, but with no further detections of these eels after 20 February 2024 (Table 2,

S1; Fig. S2). Individual eels were detected on up to 12 marine receivers (Table 2), with the maximum linear distance covered by an individual eel being approximately 104 km (including river emigration; Table 2). There was a single eel that entered Poole Harbour during its westward movement, where it stayed for approximately 20 days before moving once more.

Unlike in the river, eel detections at new receivers in the sea occurred consistently across the 24-h cycle (Fig. 4), with detection depths in the daytime down to 20 m (and always >10 m between 12.00 and 15.00h) but at as shallow as 1 m at night (Fig. 5). At sea, progression rates ranged up to 1.16 m s^{-1} , but with the majority of rates being $< 0.5 \text{ m s}^{-1}$ (Table 2).

Discussion

Integral to understanding the emigration of European silver eels back to their spawning areas in the Atlantic Ocean is generating knowledge on their initial movements out of freshwater systems and through near-shore areas, especially as this is often a two-step process, involving a period of coastal residency (Aarestrup et al., 2008) or even brief returns to fresh

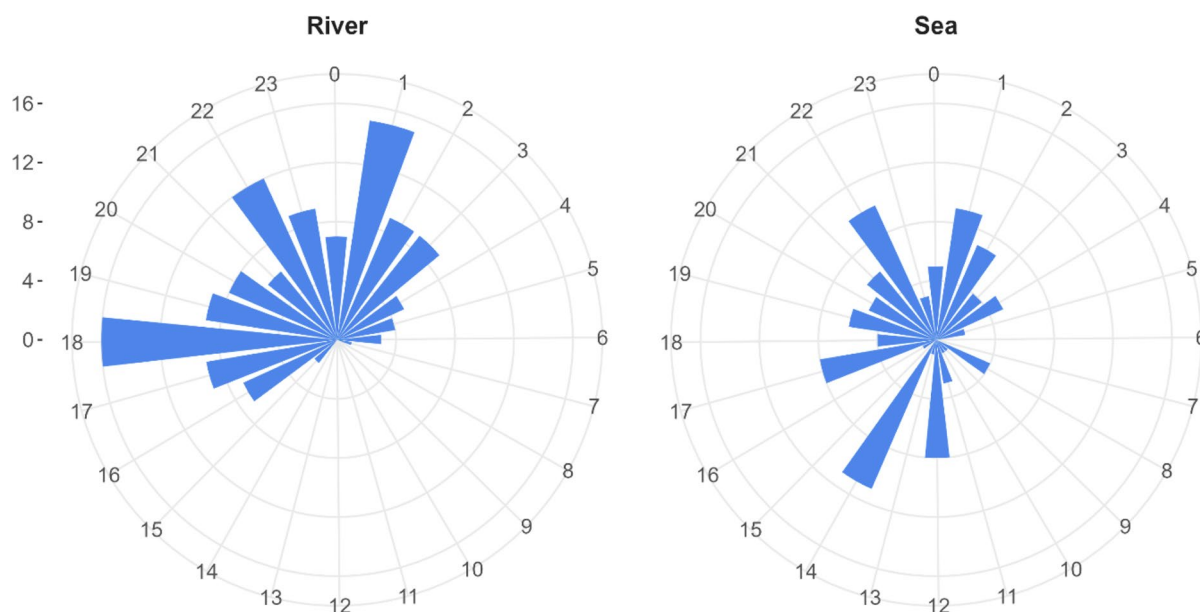


Fig. 4 Eel arrivals at downstream receivers in both the river and the sea during the 24-h cycle. The dashes on the left represent the number of observations. Data were pooled from all 25 eels

water (Tambets et al., 2021). Here, all detected tagged silver eels were moving downstream from their riverine release site into the tidal river, with their downstream movements primarily occurring at night and with their initial downstream movements coinciding with a new moon. Once in the sea, the general pattern was a westward movement of eels, although two eels initially moved eastwards.

There were 19 of the 25 tagged eels detected as successfully moving through the River Test and detected on the lower river (76% success). This success rate is in the upper range of rates reported from other studies that involved similar numbers of tagged eels, where rates have been reported as low as 15% (Verbiest et al., 2012), but can also be over 80% (Davidsen et al., 2011), with 100% success reported in some studies (Verhelst et al., 2025). The relatively high rate for the River Test might be at least in part due to the relatively short distance the eels had to migrate from their release site to the river mouth (approximately 25 km). In rivers where low silver eel migration success occurs, there is often high mortality from exploitation (e.g. Aastrup et al., 2008), hydropower and pumping station impacts (e.g. Winter et al., 2006; van Rijn et al., 2026) and predation (e.g. Radinger et al., 2025). In some cases, the emigration

might be incomplete due to the eels exhibiting a more complex, multi-step migration pattern, with individuals remaining in the lower river sections for extended periods (e.g. Tambets et al., 2021). The wider implications of the River Test results here might have been constrained slightly by the relatively modest sample size, although Cittadino et al., (2026) revealed 80% of movement variability in over 170 tagged common bream *Abramis brama* (Linnaeus, 1758) was captured in approximately 20 randomly selected fish. Moreover, the recent meta-analysis of Verhelst et al., (2025), based on tracking silver eel emigration from 18 European waterbodies, had sample sizes as low as 5 eels and as high as 895, and with highly variable silver eel migration success rates recorded (0–100%).

During their riverine stage of the migration here, the silver eels progressed downstream at rates up to 0.71 m s^{-1} , which is consistent with rates recorded in other studies (e.g. Simon et al., 2012; Frankowski et al., 2019; Merk et al., 2023). This rate was, however, higher than in the meta-analysis of Verhelst et al. (2025) that reported a mean ($\pm$ SD) rate in non-tidal reaches of $0.15 \pm 0.25 \text{ m s}^{-1}$ and in tidal reaches of $0.45 \pm 0.93 \text{ m s}^{-1}$, where the SD indicates high variability around mean values. Although river flow was not a significant influence on the downstream

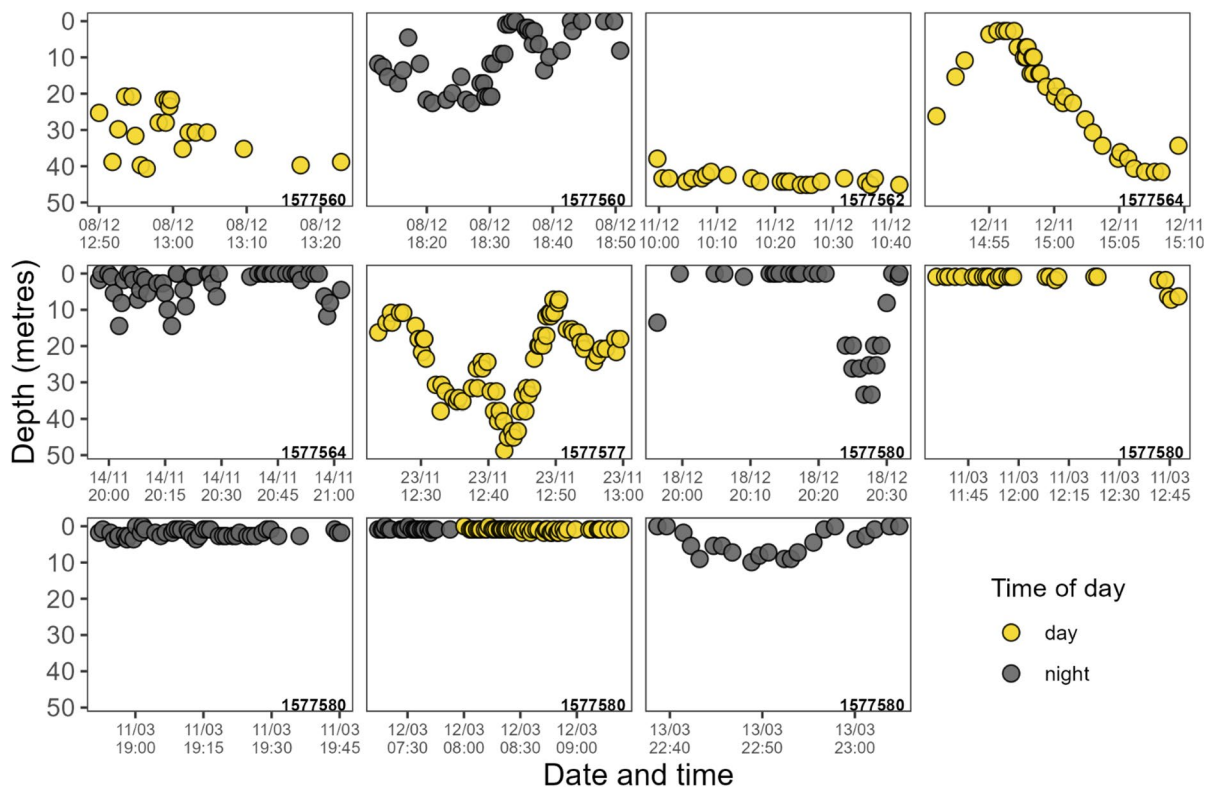


Fig. 5 Movement of eels that emigrated to sea, showing diving depths between November 2023 and March 2024. Circles show each detection with colours corresponding either to day-time (8 am–4 pm) or night. To permit visualisation, these plots

were created by filtering the data to contain a minimum of 20 sequential detections of the eel that were within four hours of the last detection. The transmitter ID corresponding to a unique eel is shown in the bottom right of each plot

progression rate here, the tagged silver eels were moving downstream in a period of relatively high river flows and levels. These flows were considered to have facilitated their downstream progression as silver eels generally use a combination of passive and active swimming during emigration (Drouineau et al., 2017; Merk et al., 2023), including selective tidal stream transport when they move during the ebb tide (Verhelst et al., 2018).

The peak timing of silver eels arriving at sea spans the period of October to January, with eels in northern Europe arriving at sea earlier than those in the south (Verhelst et al., 2025). The motivations for silver eels to migrate vary across their range, being influenced by a wide range of biological and environmental factors, plus their interactions (Vøllestad et al., 1994; Travade et al., 2010). Biological factors include the fat accumulation and lipid levels of individual eels that affect gonad development and survival during

emigration (Svedäng et al., 1997; Parzanini et al., 2021). Environmental factors include elevated river flows, changes in barometric pressure and lunar phase (Barry et al., 2016; Sandlund et al., 2017; Monteiro et al., 2020). In Ireland and Norway, the number of migrating silver eels was consistently reduced under a strong moon and increased during high and increasing water levels, most likely as these conditions reduced predation risk (Sandlund et al., 2017). Although the River Test silver eels did not show a strong response to river flow, reduced lunar illumination was the most important environmental factor influencing their downstream progression. This contrasts to Behrmann-Godel and Eckmann (2003) who found migration triggers were the first flood event that followed the full moon. Although in-river barriers did not delay the migration of silver eels in the River Test, this was considered to be due to the period of elevated flows and levels across the entire study period, as prolonged

delays to silver eel migration at barriers tends to occur during low flow periods when river levels are reduced (Piper et al., 2013). Notwithstanding, it was also considered that the motivations of the Test eels to move downstream would be inherently high. This was because these eels had already initiated their downstream movement when they were captured by the eel fishermen across a period of several weeks. The eels were then all tagged at the same time and released back into the river in early November, a period likely to be coincident with the peak migration period in this region (Verhelst et al., 2025). There was also no evidence to suggest that this brief period of captivity that imposed a migration delay resulted in eels not emigrating to sea as all detections occurred downstream, with no movements recorded in an upstream direction and 20 of the eels being detected downstream in the first five days post-release.

The marine migration routes of silver eels from northern Europe can either be through the North Sea (including around the British Isles) or be through the English Channel (Verhelst et al., 2022). Our eels entered the English Channel directly from the river and most then moved westwards. Generally, during their spawning migration, eels orient themselves westwards once they reach the continental shelf (Aarestrup et al., 2008), with one theory suggesting they retrace a migratory route imprinted during their recruitment as glass eels (Durif et al., 2022). Eels were observed swimming close to the surface at night, as initially reported in the Baltic by Westerberg et al., (2007), and while during the day they descended to depths up to 50 m, possibly to evade predation or rest (Westerberg, 1979). One eel did develop residency in Chichester Harbour for the remainder of its transmitter life, and one spent 20 days within Poole Harbour, suggesting that some individuals might use more complex migration patterns, such as those outlined by Tambets et al. (2021) where some silver eels even moved back into freshwater for short periods.

The limitations of the study have already been suggested as including a sample size of 25, with other potential issues including their holding period between their capture and tagging/ release, and the likely high migration motivations of the released tagged eels meaning their emigration cues might be less apparent than for eels tagged earlier in the year. The results indicated that only one eel was never detected, with five not being detected in the lower

river. Although these eels might have been predated or elected not to migrate further downstream, there was a chance that the transmitter was regurgitated, given gastric tagging was used. Indeed, Cottrill et al. (2006) reported 28% of gastrically tagged American silver eels *Anguilla rostrata* (Lesueur, 1817) regurgitated their tags. Also, a potential issue affecting eel detections in the study was the change in pulse rate from 30 to 50s, which was programmed to occur 20 days after tag initiation, although this was considered a minor issue given the delay was still below 60s, and so no adjustments were made to the dataset to account for the change. The use of acoustic telemetry for tracking marine movements meant that it was only possible to detect the tagged eels when they were within range of a receiver, with these understandably providing limited spatial coverage. Accordingly, should more comprehensive assessments be required of silver eel movements as they commence their marine emigration, then pop-up archival satellite tags (PSATs) could be applied. PSATs do not require in-water hardware for data collection but instead record data on a continual basis, which is then relayed to satellites when the tag automatically detaches from the fish and floats to the surface (Wright et al., 2022). Although PSATs thus provide more comprehensive data than acoustic tags, they only record data such as temperature and depth; thus, migration tracks then must be predicted using these data within, for example, geolocation and environmental matching models.

The silver eels in this chalk stream were tracked through the river using acoustic telemetry, with 76% being detected in the lower river reaches. These results suggest that silver eel migration success from this river was relatively high, despite the presence of barriers in the system, with passage likely facilitated by sufficient river flows that enabled eels to move around these barriers with minimal impediment. From a management perspective, these results emphasise the importance of maintaining river flows during migration periods (e.g. by minimising permitted abstraction where possible), ensuring barriers enable passage in both directions and, where hydropower schemes are installed on barriers, ensure that eel passage is facilitated and eel mortality minimised (Sigourney et al., 2015; Trancart et al. 2020). Nevertheless, further management and conservation insights into the factors affecting their emigration were limited due to the short time between tagging/

release and migration initiation, a likely result of high migration motivation. However, some of the eels were then detected in the marine environment, where the majority moved westwards, with most leaving the area covered by the receiver array well before their transmitters expired, thus there was little evidence to suggest these westward moving eels took up any marine residency.

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Author contributions TB, AM, JRB, SM, ES and RW conceived the study. Field data collection was carried out by TB, MI, PD, AH, AM and TS. Data analysis was performed by SM, TB, TM and JRB. SM, TB and JRB wrote the manuscript, and all authors contributed to editing the manuscript. All authors approve manuscript submission.

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Data availability The tracking data were made available to the European Tracking Network database (<http://www.lifewatch.be/etn/>), developed by the Flanders Marine Institute (VLIZ). Any other raw data used in the manuscript that are not already provided are available from the corresponding author at reasonable request.

Declarations

Conflict of interest The authors have no conflict of interest to declare that are relevant to the content of this article.

Ethical approval All aspects of data collection were completed following ethical approval and were licenced under the U.K. Animals (Scientific Procedures) Act (1986) (project licence number: PPL P3BF61EDF) under the Animals (Scientific Procedures) Act 1986).

Consent to participate This is not applicable to this manuscript.

Consent for publication This is not applicable to this manuscript.

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